

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 2 teacher notes

Sequencing note

This unit is deliberately concrete before abstract. Topic 2.1 names the parts you can point at; Topic 2.2 explains the invisible behaviour that separates them (why saving matters); Topic 2.3 moves from hardware to instructions; Topic 2.4 connects many machines. If a cohort is weak, spend the extra session on 2.1 with real equipment in hand - the later topics reuse its vocabulary constantly.

Common misconceptions to probe

1. **"The monitor is the computer."** The oldest one in the book. Probe by asking what the screen alone can do with the power on; then follow the cables. Name the system unit as the computer and the screen as its output device.
2. **"Memory and storage are the same thing."** The most consequential one for daily work. Probe with the power-cut story from Topic 2.2 before explaining; ask where an unsaved document lives, and make the group say "in RAM, and RAM forgets".
3. **"Software means only the programs you buy."** Probe with the boot-sequence question: who is in charge before any app opens? Lead them to the operating system as software.
4. **"The internet comes from the computer itself."** Probe by asking where the internet physically enters the building. Trace it to the router and the line; the lab's blinking boxes become real parts with names.
5. **"Daily app use equals machine knowledge."** Cite the Wallace and Clariana (2005) finding (average 60 percent on these concepts) and let trainees predict their own score before the unit's check questions.

Managing the activities

Topic 2.1 - Sort the donation (20 minutes, pairs or threes). The trap is arguing by brand or price. Insist the justification names the job. The two dual-role items (interactive whiteboard, barcode reader if treated as input) are the point of the activity; do not resolve them too early.

Topic 2.2 - The power-cut drill (15 minutes, pairs). The trap is vagueness ("it is lost"). Require the hierarchy words: RAM, ROM, disk, cloud. The cloud-drive scenario usually surfaces the best discussion - saved to the cloud means the local power cut does not touch the copy.

Topic 2.3 - Software census (20 minutes, groups of three). The trap is counting only visible icons. Push for the settings screen and the installed-apps list, where the system software lives. The two-question test (runs the machine vs does a task) is the takeaway; have groups write it on the board and compare wordings.

Topic 2.4 - Map the lab's network (20 minutes, pairs). If a real network exists, walk it; the physical walk is worth more than the sketch. The diagnostic sentence at the end is the assessment of the activity: it forces the cause-and-part connection.

Working with the guide's strategies

This unit is the course's most natural home for the guide's **demonstrations** and **hands-on computer exercises**. Demonstrate with the real thing wherever possible: open a case or point at the ports, pull out a RAM stick and a flash drive and hold them side by side, press save and then kill the power on a spare document (nothing teaches volatility faster). **Lectures** carry the two hierarchies (memory, software); keep them to the figures. **Interactive discussion** fits the misconception probes. **Question-answer sessions** work well at the end of each topic using the Check your understanding items as rapid rounds. **Case study analysis** is Topic 2.4's diagnostic sentence and the ERQ scenarios: treat "the internet is down on computer 14" as a case to reason through aloud.

Practical work and practicum links

The guide's practical work list maps onto this unit as follows.

- **Hands-on computer exercises:** this unit is the hands-on core of the course. Every trainee should, during the unit: identify the ports on a real machine, save a file to two different storage devices, open the installed-apps list and classify what they find, and (if any network exists) point at the switch and the router.
- **Individual assignments:** the digital-corner plan (Topic 2.1) and the spending evaluation (Topic 2.2) are individual.
- **Group assignments:** the donation sort, the software census and the network map are group tasks.
- **Internet research tasks:** the device prices for the digital-corner plan can be a short guided search; require a source for every price.
- **Presentations:** the staff-meeting explanation (Topic 2.3) is a ready-made short presentation; the network map (Topic 2.4) presents well on paper on the wall.

The practicum tasks (device inventory, RAM and storage numbers, OS and top applications, network sketch) produce a school ICT profile that Unit 4's security audit and Unit 6's emerging-technology briefing will reuse. Set them up before practicum and debrief after.

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